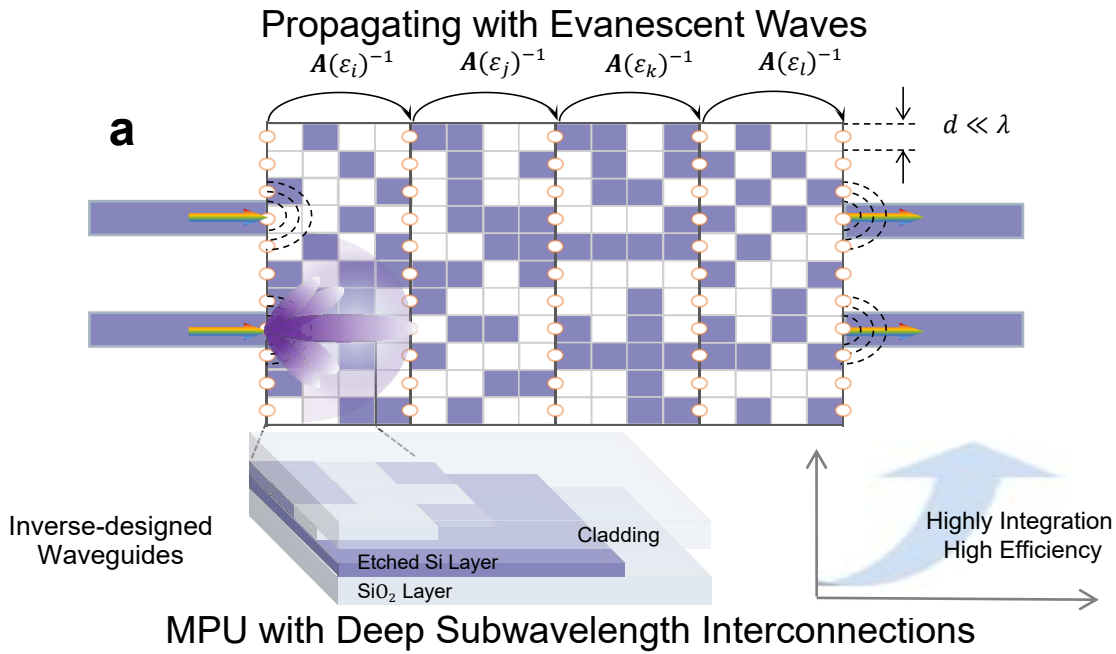
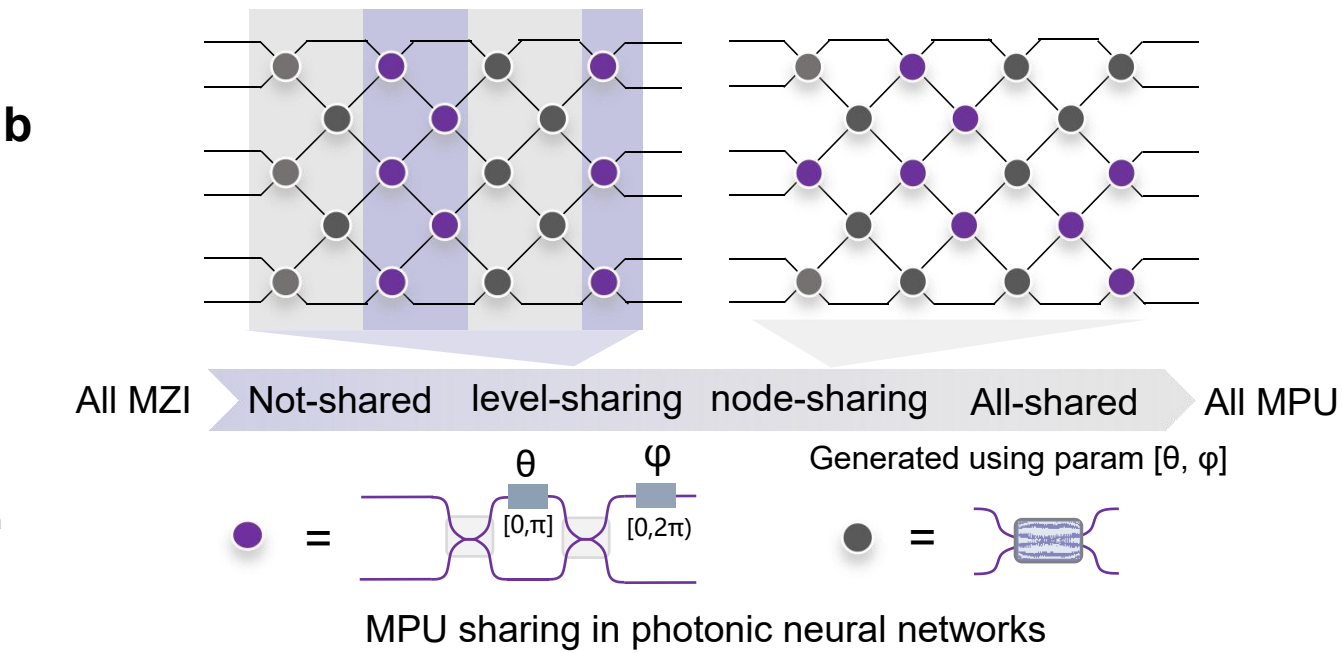
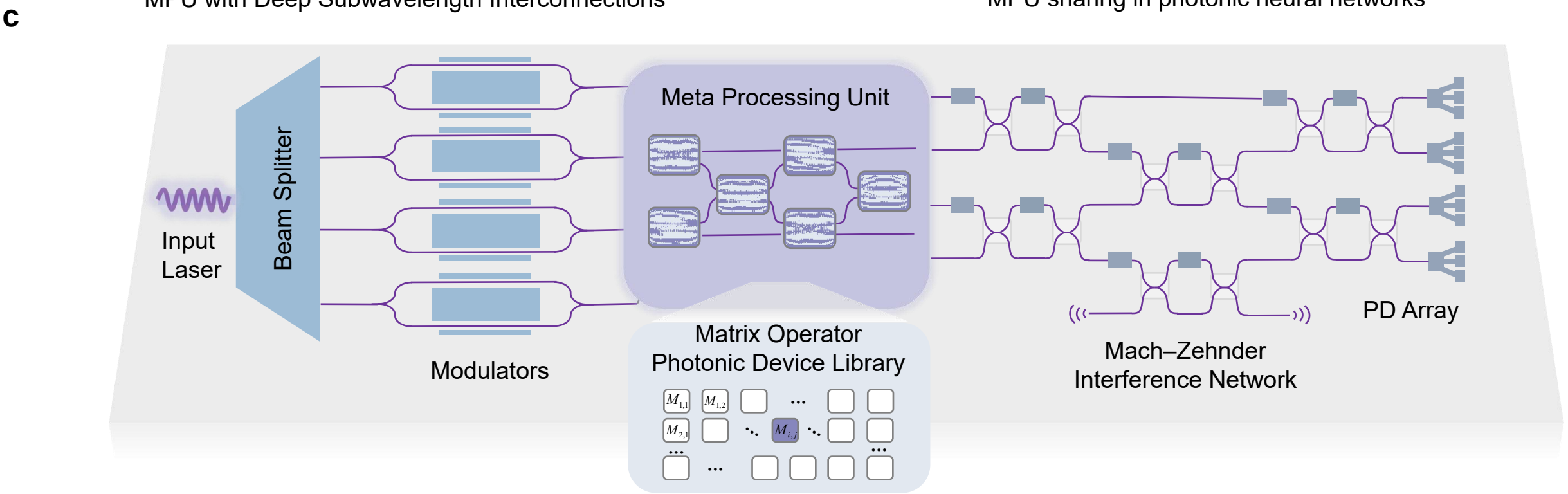

Propagating with Evanescent Waves
$A(\varepsilon_i)^{-1}$
$A(\varepsilon_j)^{-1}$
$A(\varepsilon_k)^{-1}$
$A(\varepsilon_l)^{-1}$
a
$d \ll \lambda$
Inverse-designed Waveguides
Cladding
Etched Si Layer
SiO_2 Layer
Highly Integration
High Efficiency
MPU with Deep Subwavelength Interconnections


b
All MZI
Not-shared
level-sharing
node-sharing
All-shared
All MPU
θ
φ
[0,π]
[0,2π)
Generated using param [θ, φ]
=
=
MPU sharing in photonic neural networks


c
Input Laser
Beam Splitter
Modulators
Meta Processing Unit
Matrix Operator
Photonic Device Library
Mach–Zehnder
Interference Network
PD Array